Array

1-D (Vector)

$$\begin{bmatrix} 1 \\ 5 \\ 0 \end{bmatrix}$$

3×1

2-D (Matrix)

$$2\text{-D} \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 9 \end{bmatrix}$$

2×3

3-D

Dot Product

Condition : No of columns of first array = No of rows of second array

Shape : No of rows of first array * No of columns of second array

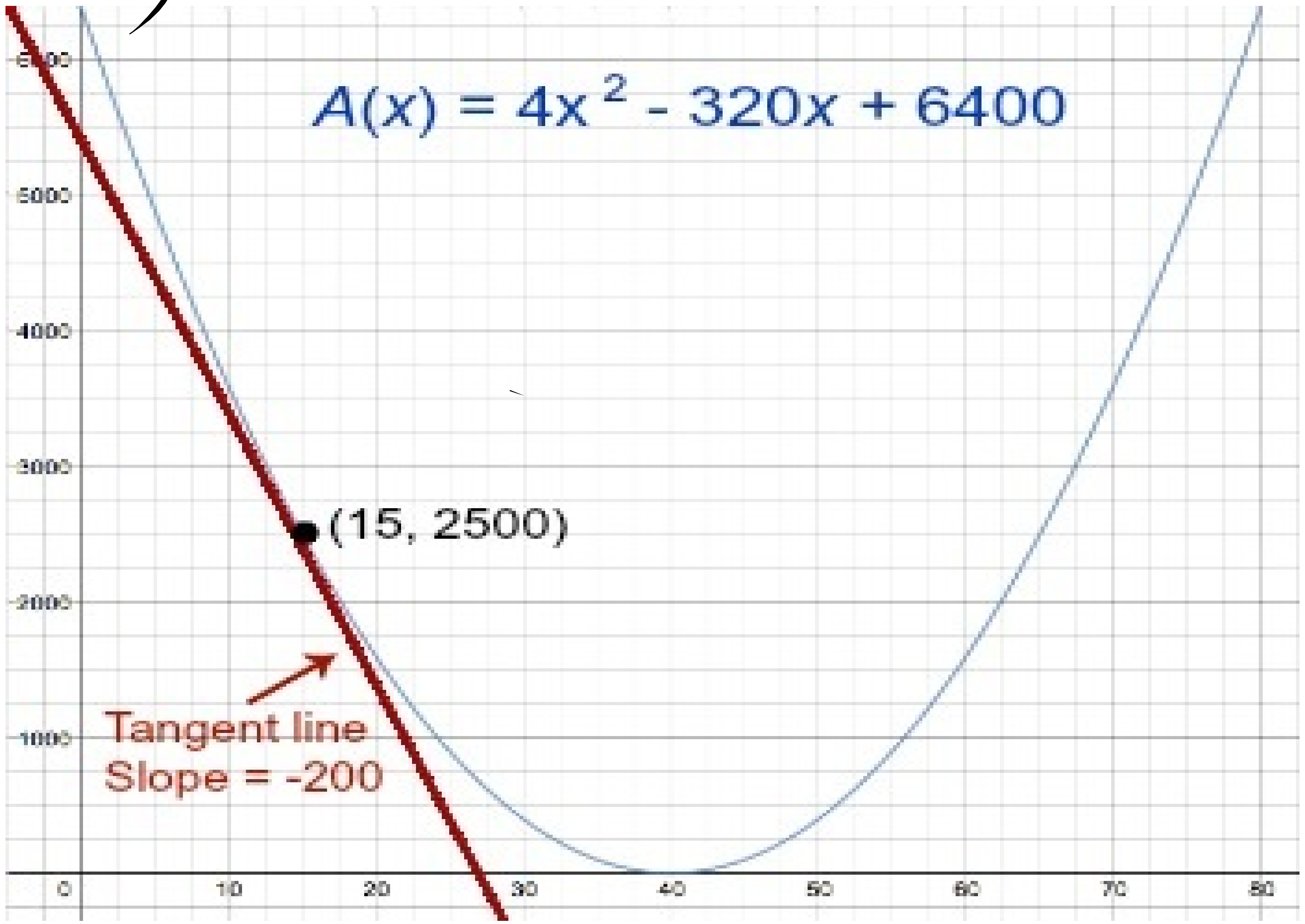
$$y = 3x^2 + 5x + 10$$

$$y' = 6x^1 + 5 + 0$$

$$y = 4x^2 - 320x + 6400$$

$$f_{y'}(x) = 8x - 320$$

$$y'' = 8$$



$$y = -0.05x^2 + 2x - 15$$

$$y' = -0.1x + 2$$

$$y'' = -0.1$$

$$y = 3x^4 - 2x^3 + 4x^2 - 5x + 1.$$

$\frac{d}{dx}$ $y' = 12x^3 - 6x^2 + 8x - 5$

$\frac{d}{dx}$ $y'' = 36x^2 - 12x + 8$

$z = x^2 + y^2$

$f(x,y) = x^2 + y^2$

Partial Derivative

$$Z = x^2 + y^2$$

$$\frac{\partial Z}{\partial x} = 2x + 0$$

$$\frac{\partial Z}{\partial y} = 0 + 2y$$

Chain Rule

$f(x) = (g(x) - 2)^3, g(x) = 5x^4.$

1- $y = (g-2)^3$

2- $g = 5x^4$

$$\frac{dy}{dg} = 3(g-2)^2 \times 1$$

$$\frac{dy}{dx} = \frac{dy}{\cancel{dg}} \frac{\cancel{dg}}{dx}$$

$$\frac{dg}{dx} = 20x^3$$

$$\frac{dy}{dx} = 3(g-2)^2 \cdot 20x^3 = 3(5x^4-2)^2 \cdot 20x^3$$

If z = 1/3 y and y = x² + 3x⁴ + 10 what is

$$z = \frac{1}{3}y^2 \qquad y = x^2 + 3x^4 + 10 \qquad \frac{dz}{dx} = ?$$

$$\frac{dz}{dy} = \frac{2}{3}y \;,\quad \frac{dy}{dx} = 2x + 12x^3 \qquad \frac{dz}{\cancel{dy}} \cdot \frac{\cancel{dy}}{dx}$$

$$\frac{dz}{dx} = \frac{2}{3}y \cdot 2x + 12x^3$$

$$= \left[\frac{2}{3}\left(x^2 + 3x^4 + 10\right)\right] \cdot 2x + 12x^3$$

$$\left[\frac{2}{3}x^2 + 2x^4 + \frac{20}{3}\right]\left[2x + 12x^3\right]$$

Recap

Array

1-D(vector)

2-D(matrix)

element wise operations

Rate of change

Slope

$\frac{\Delta y}{\Delta x}$

y' (first derivative)

$y = (2x^2 + 5x - 1)^2 * 4x + 5$

$y' = 2(2x^2 + 5x - 1) * 4$

partial derivative

$y = x^2 + y^3$

$\frac{\partial y}{\partial x} = 2x + 0$

Chain Rule